

UNIT - II

11R - digital filter design

Analog filter

Digital filter

i) structure  Resistor

i) summer, multiplier,
delay element

A diagram showing a curved arrow pointing from (i) to (ii) with the label LT.

i.) ZT $\rightarrow$ digital Tr-h

$$H(s) = \frac{Y(s)}{X(s)}$$

analogy T.M.

$$H(z) = \frac{Y(z)}{X(z)}$$

(ii) $h(n)$

iii) $h(t) \rightarrow$ filter cost

iv) flexibility - less

iv) Horribility → More

$$H(s)$$

Analogy Tr. In

Mapping

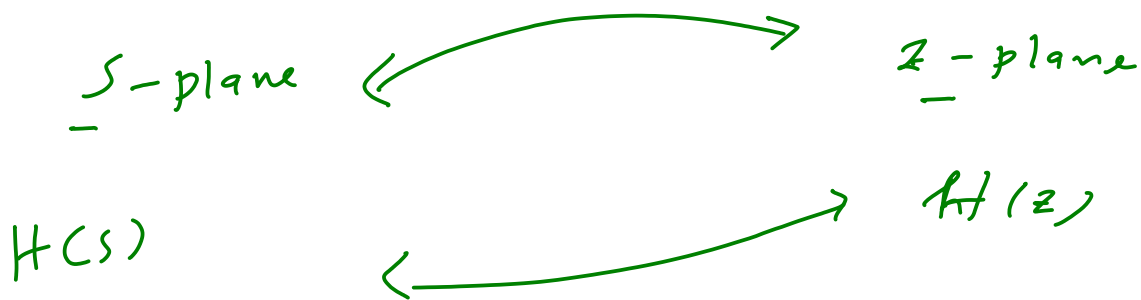
$$H(z)$$

Digital Tr-fn

Design of file ^{analog}

 $H(s)$
$$h(t)$$
$$H(z)$$
 $h(\omega)$

Design of digital filter



Relation between LT & ZT

LT $\rightarrow$ cont. signal $x(t)$

$$\text{LT}[x(t)] = X(s) = \int_{-\infty}^{\infty} x(t) e^{-st} dt \quad \text{--- (1)}$$

$$\text{LT}[x(nT)] = X(s) = \sum_{n=-\infty}^{\infty} x(nT) e^{-snT} \quad \text{--- (2)}$$

ZT $\rightarrow$ digital signal $\rightarrow x(n)$

$$\text{ZT}[x(n)] = X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \quad \text{--- (3)}$$

$$\text{eqn (2)} \Rightarrow X(s) = \sum_{n=-\infty}^{\infty} x(nT) \left(e^{sT}\right)^{-n} \quad \text{--- (4)}$$

compare (3) & (4)

$$X(s) = X(z) \quad \text{at} \quad \boxed{z = e^{sT}}$$

$z \rightarrow$ complex variable

$$z = r \cdot e^{j\omega}$$

(5)

$s \rightarrow$ complex variable

$$s = \sigma + j\omega$$

(1)

We know that

$$z = e^{sT}$$

$$\underline{r} \cdot e^{j\omega} = e^{(\sigma + j\omega)T} = \underline{e^{\sigma T}} \cdot e^{j\omega T} \quad (7)$$

$$\gamma = e^{\sigma T}$$

$$e^{j\omega} = e^{j\omega T}$$

$$\omega = \omega T$$

Analogy freq

Sampling period

Digital freq

$$\gamma = e^{\sigma T}$$

Mapping procedure between
 s -plane with z -plane

$$\underline{\sigma < 0} \rightarrow \sigma = -\infty$$

$$\gamma = e^{\sigma T} = e^{-\infty \cdot T} = 0$$

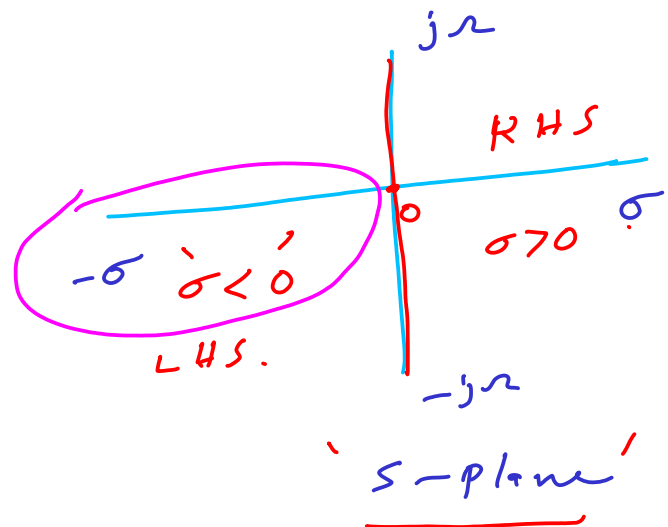
$$0$$

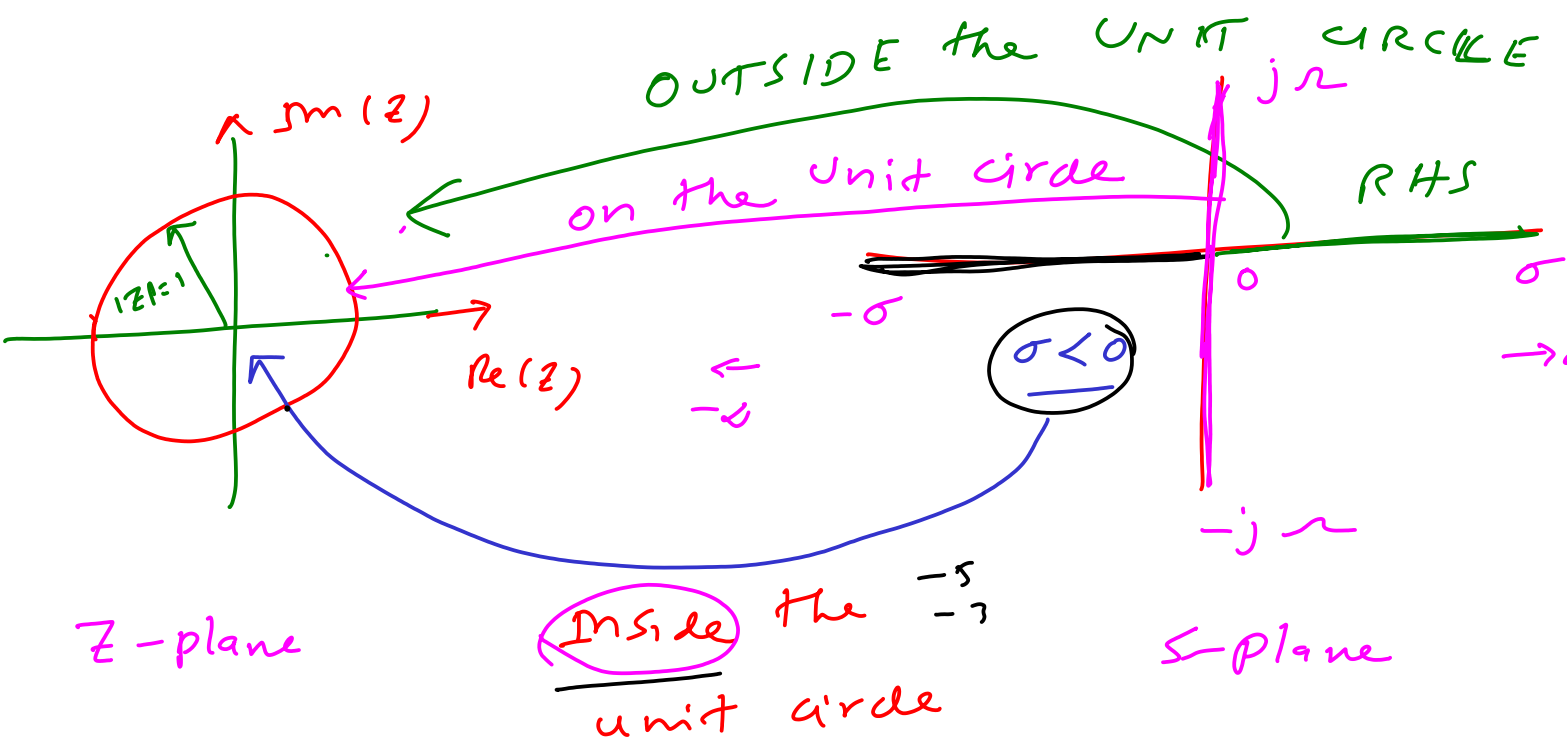
$$\underline{\sigma > 0} \rightarrow \sigma = \infty$$

$$\gamma = e^{\sigma T} = e^{\infty \cdot T} = \infty$$

$$\underline{\sigma = 0}$$

$$\gamma = e^{\sigma T} = e^0 = 1$$





s -plane $\longleftrightarrow$ z -plane
 LHS $\longrightarrow$ inside the unit circle
 RHS $\longrightarrow$ OUTSIDE the unit circle
 Imaginary axis $\longrightarrow$ on the unit circle

Design of Analog filter

" $H(s)$ " $\xrightarrow{\text{Stability}}$ $= \frac{Y(s)}{X(s)}$

'poles' $\rightarrow$ location $\leftarrow$ Stability $\rightarrow$ 'all poles' must lie in the LHS- s -plane

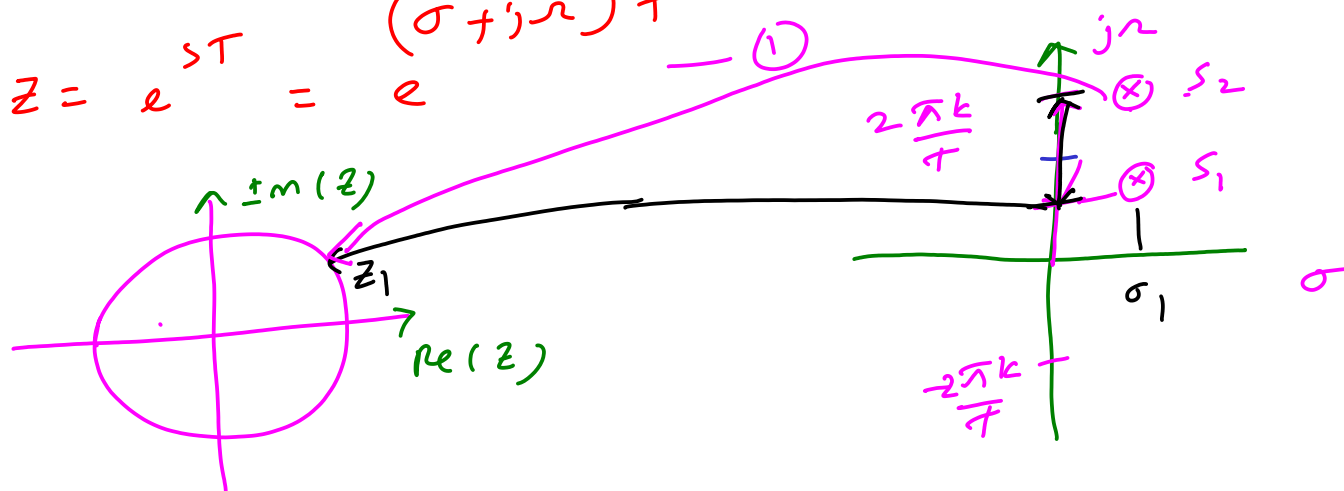
Digital filter $\rightarrow$ all poles $\rightarrow$ Must lie inside the unit circle

②

$$z = \sigma \cdot e^{j\omega}$$

$$s = \sigma + j\omega$$

$$z = e^{sT} = e^{(\sigma + j\omega)T}$$



$$s = s_1 = \sigma_1 + j\omega_1 \checkmark$$

$$z_1 = e^{s_1 T} = e^{(\sigma_1 + j\omega_1)T} = e^{\sigma_1 T} \cdot e^{j\omega_1 T} \quad \text{--- (2) } \checkmark$$

$$s = s_2 = \sigma_1 + j\left(\omega_1 + \frac{2\pi k}{T}\right) \checkmark$$

$$z_2 = e^{s_2 T} = e^{[\sigma_1 + j\omega_1 + j\frac{2\pi k}{T}]T}$$

$$= e^{\sigma_1 T} \cdot e^{j\omega_1 T} \cdot e^{j2\pi k} = e^{\sigma_1 T} \cdot e^{j\omega_1 T} \cdot e$$

$$z_2 = e^{\sigma_1 T} \cdot e^{j\omega_1 T} \quad \text{--- (3) } \checkmark$$

$$\omega \rightarrow \frac{2\pi k}{T}$$

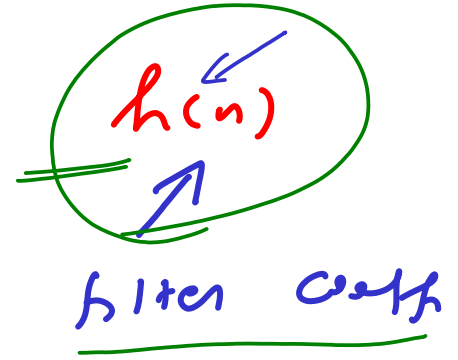
$$\begin{array}{ccc} \omega_1 & \rightarrow & \omega_1 \\ \omega_2 & \rightarrow & \omega_2 \\ \omega_3 & \rightarrow & \omega_3 \end{array}$$

multiple analog poles $\rightarrow$ single Digital pole
"ALIASING"

IIR filter ← digital

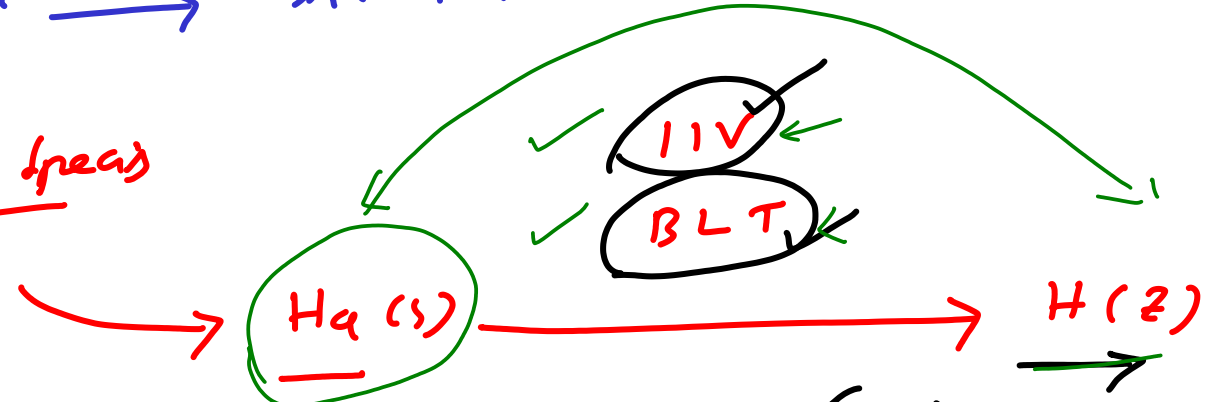
→ Infinite Impulse Response

filter



IIR digital $\xrightarrow{X}$ $h(n) \leftarrow$ infinite duration

Analog spec



Impulse Invariant Method (IIV)
Bilinear Transformation Tech (BLT)

Structure of IIR

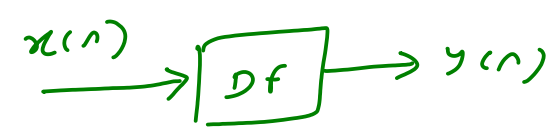
- Direct form - I ✓
- Direct form - II ✓
- Cascaded form - II ✓

LT ← Analog filter ✓

R, L, C

ZT ← Digital filter ✓

Adder, Mult, Scalar



DF - IIR $\Rightarrow$

$$y(n) = - \sum_{k=1}^N a_k \cdot y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$N \rightarrow$ order of the filter

DIRECT FORM-I

IIR filter $\rightarrow$ recursive filter
 feedback filter

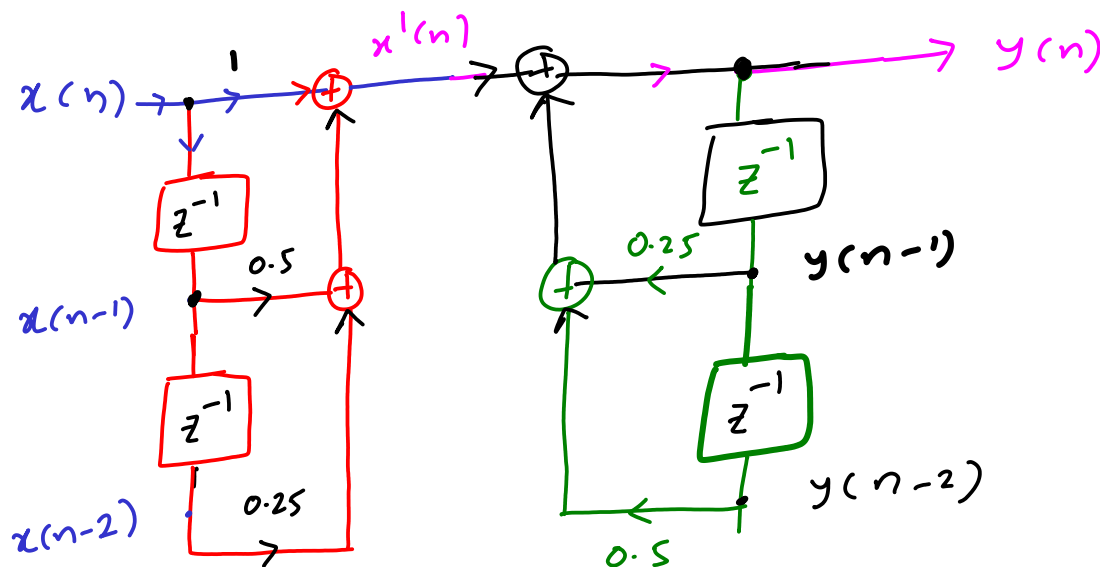
$$y(n) = - \sum_{k=1}^N a_k y(n-k) + \sum_{k=0}^M b_k x(n-k)$$

$y(n) \rightarrow$ fn [i/p, past o/p]

Time domain Relationship bwn i/p & o/p

- ① obtain Direct form-I Realization of a digital IIR filter described by the following difference equation

$$y(n) = 0.25 y(n-1) + 0.5 y(n-2) + 0.25 x(n-2) + 0.5 x(n-1) + x(n)$$



$z^{-1} \rightarrow$ delay element

④

(2) obtain Direct form I realization of a digital IIR filter with a digital transfer function

$$H(z) = \frac{1 + 0.5z^{-1} + 0.25z^{-2}}{0.25 + 0.5z^{-1} + 0.6z^{-2} + z^{-3}}$$

$$\frac{Y(z)}{X(z)} = \frac{1 + 0.5z^{-1} + 0.25z^{-2}}{0.25 + 0.5z^{-1} + 0.6z^{-2} + z^{-3}}$$

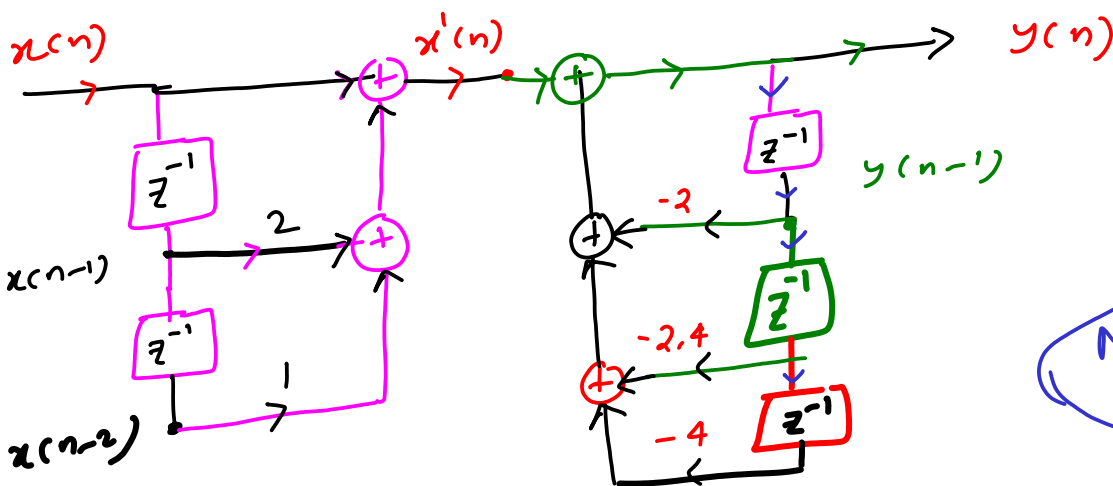
$$Y(z) \cdot 0.25 + 0.5z^{-1}Y(z) + 0.6z^{-2}Y(z) + z^{-3}Y(z) = X(z) + 0.5z^{-1}X(z) + 0.25z^{-2}X(z)$$

(I)

$$z^{-1} [x(n-k)] = z^{-k} X(z)$$

take z^{-1} of eqn (I)

$$y(n) = -2y(n-1) - 2.4y(n-2) - 4y(n-3) + 4x(n) + 2x(n-1) + 1 \cdot x(n-2) \quad \text{--- (II)}$$



5 - delay elements
Memory elements

DIRECT FORM - I

① $I \pm R \rightarrow H(z) = \frac{Y(z)}{X(z)} \rightarrow NR \rightarrow \text{only (ve) powers of 'z'}$
 $\rightarrow dr$

$$y(n) = -2y(n-1) - 2.4y(n-2) - 4y(n-3) + 4x(n) + 2x(n-1) + x(n-2)$$

digital filter,
difference eqn

←
differential eqn $\rightarrow$ Analog filter

DIRECT form - II ?

$$H(z) = \frac{Y(z)}{X(z)}$$

② $H(z) = \frac{Y(z)}{X(z)} = \frac{Y(z)}{W(z)} \cdot \frac{W(z)}{X(z)}$

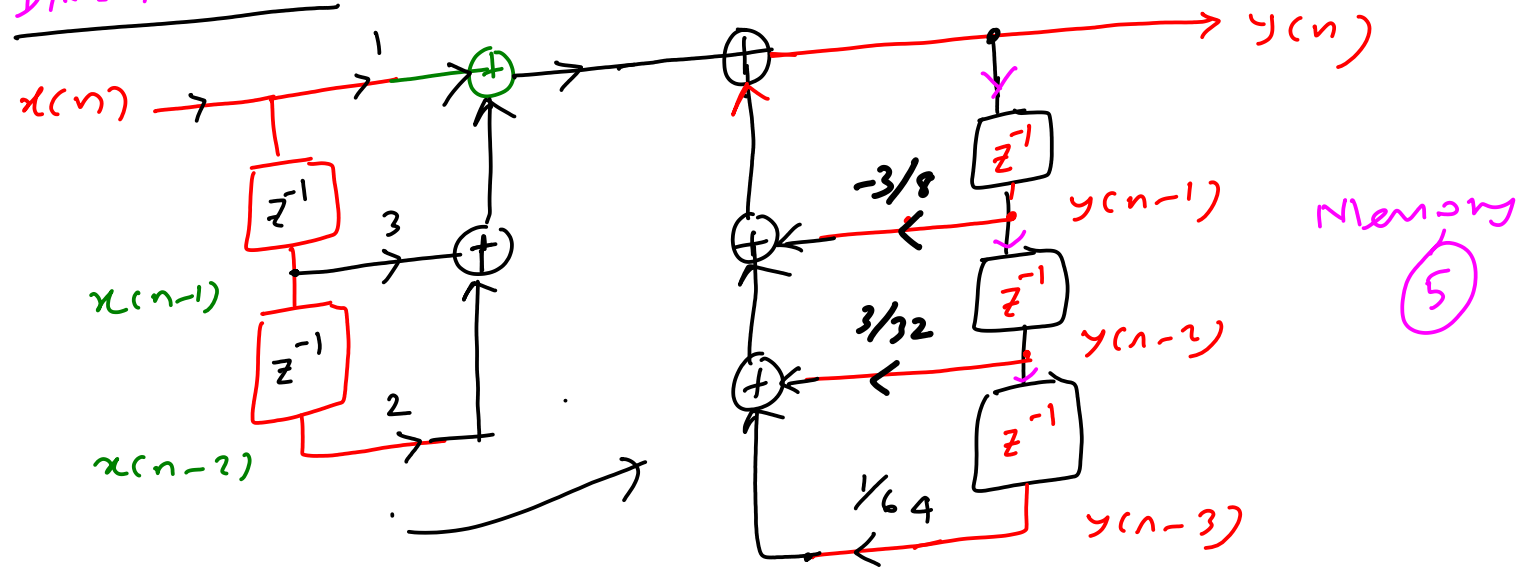
③ $\frac{W(z)}{X(z)} = \frac{1}{Dr. \text{ polynomial}} \leftarrow \text{structure}$

④ $\frac{Y(z)}{W(z)} = Nr. \text{ polynomial} \leftarrow \text{structure}$

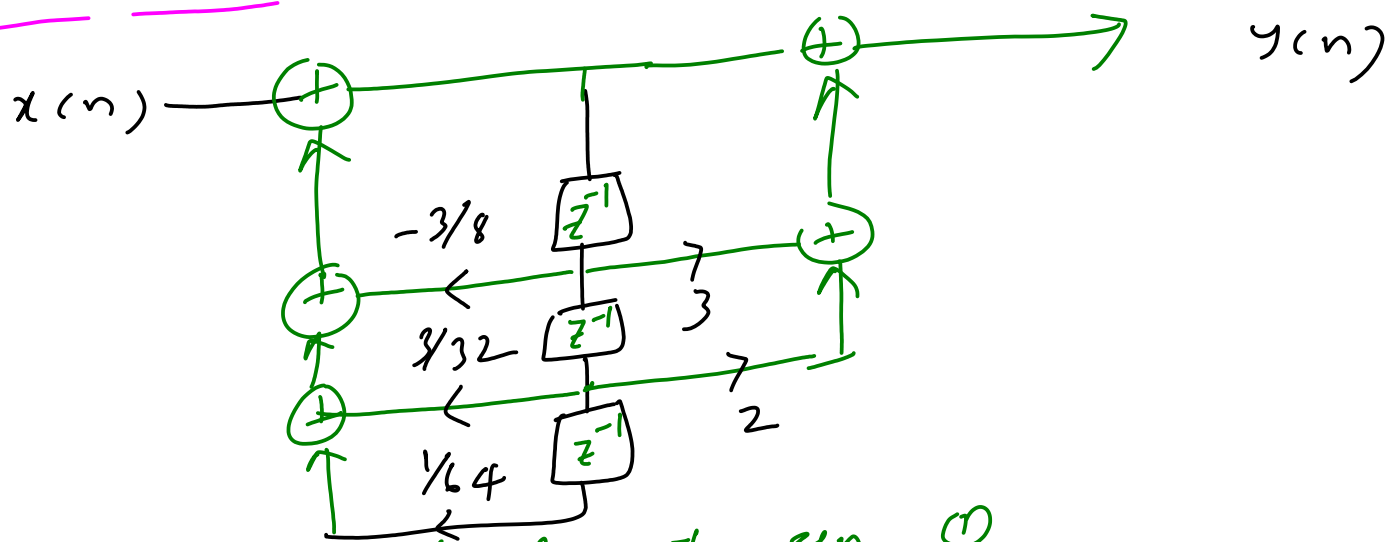
⑤ combine str 1 & str 2 $\rightarrow$ overall structure

- ② Obtain the Direct form I, direct form-II realisations of the LTI system governed by the equation
- $$y(n) = -\frac{3}{8} y(n-1) + \frac{3}{32} y(n-2) + \frac{1}{64} y(n-3) + x(n) + 3x(n-1) + 2x(n-2)$$

DIRECT FORM-I



DIRECT FORM-II



Apply ZT on both sides of eqn ①

$$Y(z) = -\frac{3}{8} z^{-1} Y(z) + \frac{3}{32} z^{-2} Y(z) + \frac{1}{64} z^{-3} Y(z) + X(z) + 3z^{-1} X(z) + 2z^{-2} X(z)$$

$$Y(z) + \frac{3}{8} z^{-1} Y(z) - \frac{3}{32} z^{-2} Y(z) - \frac{1}{64} z^{-3} Y(z) = X(z) + 3z^{-1} X(z) + 2z^{-2} X(z)$$

$$Y(z) \left[1 + \frac{3}{8} z^{-1} - \frac{3}{32} z^{-2} - \frac{1}{64} z^{-3} \right] = X(z) \left[1 + 3z^{-1} + 2z^{-2} \right]$$

$$\therefore \frac{Y(z)}{X(z)} = H(z) = \frac{1 + 3z^{-1} + 2z^{-2}}{1 + \frac{3}{8} z^{-1} - \frac{3}{32} z^{-2} - \frac{1}{64} z^{-3}} \quad \text{--- (i)}$$

$$\frac{Y(z)}{X(z)} = \frac{Y(z)}{W(z)} \cdot \frac{W(z)}{X(z)} \quad \text{--- (ii)}$$

$$\frac{W(z)}{X(z)} = \frac{1}{1 + \frac{3}{8} z^{-1} - \frac{3}{32} z^{-2} - \frac{1}{64} z^{-3}} \quad \text{--- (iv)}$$

$$\boxed{\frac{Y(z)}{W(z)} = 1 + 3z^{-1} + 2z^{-2}} \quad \text{--- (v)}$$

Consider eqn (4)

$$W(z) + \frac{3}{8} z^{-1} W(z) - \frac{3}{32} z^{-2} W(z) - \frac{1}{64} z^{-3} W(z) = X(z) \quad \text{--- (vi)}$$

Take z^{-1} of eqn (vi)

$$W(n) + \frac{3}{8} W(n-1) - \frac{3}{32} W(n-2) - \frac{1}{64} W(n-3) = X(n)$$

$$\therefore \checkmark W(n) = X(n) - \frac{3}{8} W(n-1) + \frac{3}{32} W(n-2) + \frac{1}{64} W(n-3) \quad \text{--- (vii)}$$

Consider eqn (v)

Take z^{-1} of eqn (v)

$$\frac{Y(z)}{W(z)} = 1 + 3z^{-1} + 2z^{-2}$$

$$Y(z) = \underline{W(z)} + 3 \underline{z^{-1} W(z)} + 2 \underline{z^{-2} W(z)} \quad \text{--- (viii)}$$

Take z^{-1} of eq (viii)

$$y(n) = w(n) + 3w(n-1) + 2w(n-2) \quad \text{--- (ix)}$$

$\therefore$ Consider eq (7) & (9) to obtain the Direct form II structure of given IIR filter

